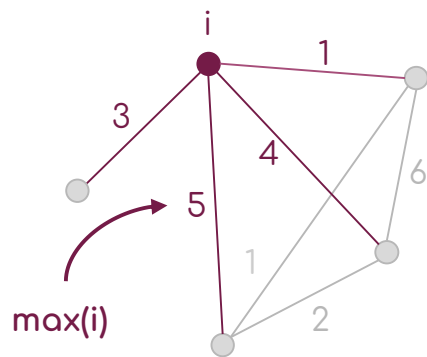
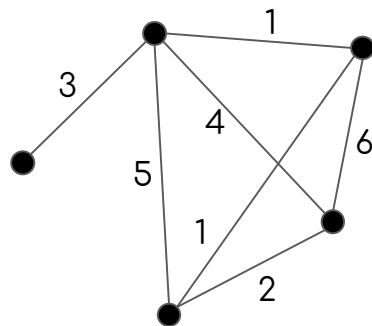
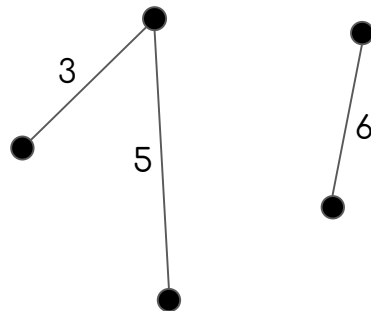


Approximating a Maximum Spanning Tree Problem

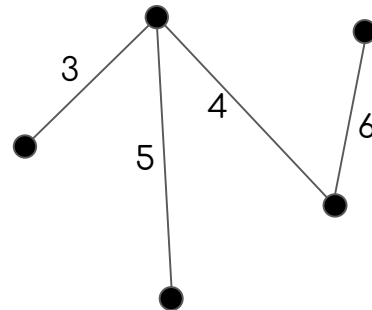
Let $G = (V, E)$ be an undirected graph with distinct edge weights $w(u, v)$ on each edge $(u, v) \in E$. For each vertex $v \in V$, let $\max(v) = \max_{(u,v) \in E} \{w(u, v)\}$ be the maximum-weight edge incident on that vertex. Let $S_G = \{\max(v) : v \in V\}$ be the set of maximum-weight edges incident on each vertex, and let T_G be the maximum-weight spanning tree of G , that is, the spanning tree of maximum total weight. For any subset of edges $E' \subseteq E$, define $w(E') = \sum_{(u,v) \in E'} w(u, v)$.



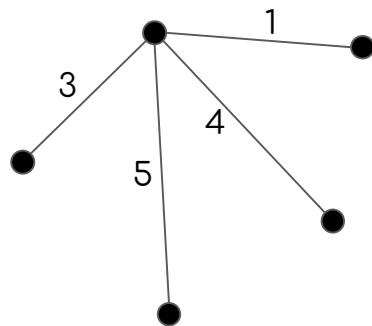
$S(G)$



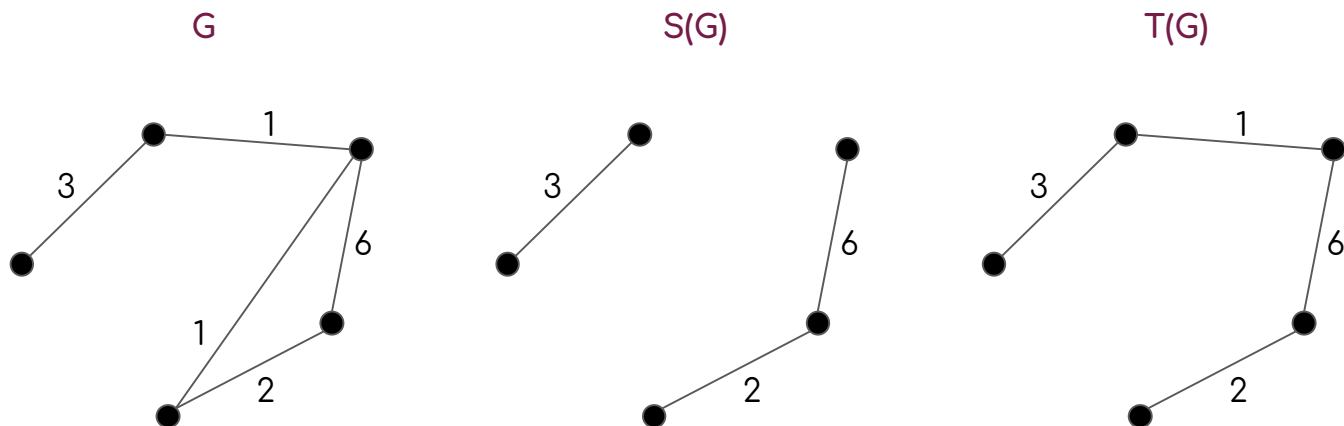
$T(G)$



a. Give an example of a graph with at least 4 vertices for which $S_G = T_G$.



b. Give an example of a graph with at least 4 vertices for which $S_G \neq T_G$.



c. Prove that $S_G \subseteq T_G$ for any graph G .

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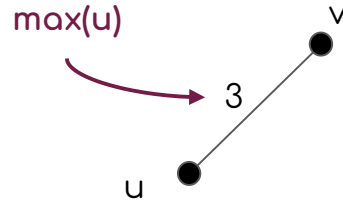
Consider Kruskal's algorithm:

Sort the edges from largest to smallest weight,

Consider them one at a time in order, and if the edge doesn't introduce a cycle, add it to the tree.

Proof by Contradiction

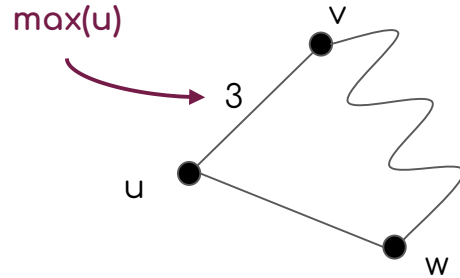
There is an edge (u, v) which is in $S(G)$ and not in $T(G)$



(u, v) is not added to the maximum spanning tree
So it must introduce a cycle

Proof by Contradiction

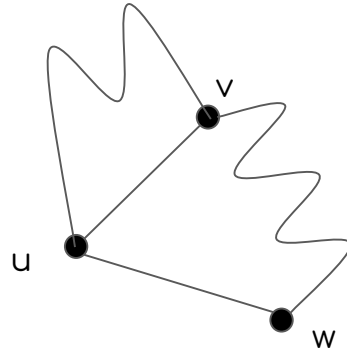
If (u, v) is introducing a cycle, there is some edge (u, w) which is already included in the spanning tree



Removing (u, w) and adding (u, v) should lead to a better maximum spanning tree.

Proof by Contradiction

If by doing that a cycle appears, there were already a path between u and v



That path + (u, w) + the path from w to v , creates a cycle in the original spanning tree

d. Prove that $w(T_G) \geq w(S_G)/2$ for any graph G .

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Based on part c. all edges in S_G are already in T_G .

Therefore $w(T_G) \geq w(S_G) \geq w(S_G)/2$

- e.*** Give an $O(V + E)$ -time algorithm to compute a 2-approximation to the maximum spanning tree.

Let $N(v)$ denote the neighbors of a vertex v . An algorithm APPROX-MAX-SPANNING that finds S_G produces a subset of edges of a minimum spanning tree by part (c), and is a 2-approximation algorithm by part (d).

Let $N(v)$ denote the neighbors of a vertex v . The following algorithm APPROX-MAX-SPANNING produces a subset of edges of a minimum spanning tree by part (c), and is a 2-approximation algorithm by part (d).

Algorithm 6 APPROX-MAX-SPANNING-TREE(V, E)

```
1:  $T = \emptyset$ 
2: for  $v \in V$  do
3:    $max = -\infty$ 
4:    $best = v$ 
5:   for  $u \in N(v)$  do
6:     if  $w(v, u) \geq max$  then
7:        $v = u$ 
8:        $max = w(v, u)$ 
9:     end if
10:  end for
11:   $T = T \cup \{v, u\}$ 
12: end for
13: return  $T$ 
```
